

Byung-cheol Kang

Bioinformatician · Bacteriophage Research · In Silico Design

PROFILE

South Korea

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Bioinformatician with more than six years of experience connecting microbial and bacteriophage genomics, metagenomics, reproducible workflow engineering, and research operations. Currently at **Microiotix**, conducting bacteriophage research, in silico design, and research database development and evaluation. Previously led a five-person bioinformatics team and built shared Snakemake workflows for 16S, ITS, WGS, and hybrid assembly.

6+ years

Applied bioinformatics

4 profiles

16S · ITS · WGS · Hybrid

98+

Archived project-order lower bound

7 papers

SCI(E) publication record

EXPERIENCE

Microiotix · Bioinformatician

2026.07 - PRESENT

- Conduct bacteriophage research with computational analysis supporting research decisions.
- Perform in silico design for bacteriophage-related research and development.
- Build and evaluate research databases for reproducible candidate assessment.

Sanigen · Bioinformatics Team Lead · Senior Researcher

2020.03 - 2025.10

- Led a five-person bioinformatics team delivering microbial WGS, metagenomics, assembly, annotation, comparative-genomics, and research support.
- Designed and maintained a shared Snakemake platform covering 16S, ITS, WGS, and Illumina-ONT hybrid assembly with validation, logging, quality gates, and report packaging.
- Standardized execution with Conda, Docker, and Apptainer across Linux compute environments; wrote Python, R, and Bash utilities for QC, analysis, visualization, and reporting.
- Supported comparative-genomics marker discovery and foodborne-pathogen NGS panel development that contributed to two 2023 papers and one joint patent.
- Coordinated analysis review and delivery across a conservative lower bound of 98 archived project orders.

GinaPath · Business Development

2019.11 - 2020.01

- Supported genomics-service requirements, project scoping, and client-facing technical communication.

SELECTED PROJECTS

PLATFORM Integrated microbial genomics platform

Unified 16S, ITS, WGS, and hybrid profiles under a shared Snakemake execution contract. Active tooling included FastQC, MultiQC, QIIME 2/ DADA2, SPAdes, Flye/Unicycler, QUASt, BUSCO, Prokka, Abriicate, eggNOG-mapper, and custom reports.

RESEARCH Comparative-genomics marker discovery

Narrowed complete-genome differences to an approximately 17 kb candidate region while separating computational prioritization from downstream assay validation.

PHAGE Bacteriophage assembly and annotation

Reviewed assembly evidence, produced single-contig final assemblies in the assessed set, and supplemented Prokka results with RAST-tk evidence and explicit completion caveats.

INDEPENDENT R&D Phage lifestyle evidence prototype

Designed an interpretable evidence workflow combining CheckV, Pharokka/Phold, BACPHLIP, and PhaTYP while routing disagreement to manual review.

TECHNICAL SKILLS

Genomics	Microbial WGS, assembly, annotation, comparative genomics, bacteriophage analysis
Workflow engineering	Snakemake, Nextflow, Conda, Docker, Apptainer, Linux
Programming	Python, R, Bash, Git
Selected tools	QIIME 2, DADA2, FastQC, MultiQC, SPAdes, Flye, Unicycler, QUASt, BUSCO, Prokka, Abriicate, eggNOG-mapper

EDUCATION

M.S., Medical Science · Kyungpook National University School of Medicine · 2016.02 - 2018.08

B.S., Biotechnology; Minor in Psychology · Kyungpook National University · 2008.03 - 2016.02

SELECTED PUBLICATIONS & IP

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- Park et al. Novel next generation sequencing panel method for multiple detection and identification of foodborne pathogens in agricultural wastewater. **Frontiers in Microbiology** 14 (2023).
 - Park et al. Development and evaluation of a next-generation sequencing panel for pathogens in fermented foods. **Journal of Microbiology and Biotechnology** 33(1) (2023).
 - Kang et al. Nothofagin suppresses mast cell-mediated allergic inflammation. **Chemico-Biological Interactions** 298 (2019).
 - Joint patent contribution: NGS-based primer set and detection method for foodborne bacteria (2022).

LANGUAGE

Korean - Native English - IELTS 6.5 (October 2023)